A Finite Black-Hole Information Ledger for Observer-Patch Holography

The OPH collaboration

October 17, 2026

Paper release: `r2050` **Released:** October 17, 2026

Collaboration affiliation: Pragma Research Inc.

Abstract

We formulate a finite information ledger for observer-patch models using machine-checked dynamics and record channels. A continuous one-parameter star-automorphism group on a finite-dimensional private algebra acts by blockwise unitary conjugation. For a supplied projective partition, the public record channel is the exact uniform mixture of independent sign unitaries. Classical relative-entropy contraction and finite fluctuation identities supply complementary thermodynamic inputs. A declared rich-fibre adapter constructs noncommutative regional algebras with commuting disjoint blocks and unique restriction gluing. We state the support-aware entropy identity and its proposed thermality and information-relocation consequences as finite conjectures. Together these results and conjectures make the black-hole information question precise within the supplied finite model. On the layered source family of the spacetime construction the number of unordered read pairs crossing a surface per layer is an exact area law, $(\pi/4)n^2a^4$ per unit area for site density n and read radius a, and reading one crossing read as one nat of cut entropy fixes the Planck-area dictionary $n^2a^4 = 1/(\pi\ell_P^2)$.

Finite-model scope and theorem boundary

Statements labeled *Theorem* are machine-checked in the Lean library at the cited modules. Statements labeled *Conjecture* are the explicitly unproved statements; nothing in this paper should be read as claiming them. The phrase "black hole" names the shape of the question, not a constructed physical object: the model has no continuum horizon, no gravitational collapse, no Hawking flux, and no evaporation endpoint. The bridge from the finite ledger to physical black-hole statements requires a faithful continuum realization of the finite source-causal carrier, a source-realized clock, Einstein-branch geometry and stress, and, for any observable, a prospectively fixed measurement contract under immutable custody. No observable prediction follows from this finite ledger.

1 The question in finite form

The black-hole information problem sets two commitments against each other: microscopic dynamics that preserves information, and a macroscopic record that looks exactly thermal. In the observer-patch model both commitments are formal objects, and the question becomes finite: given that ambient dynamics on a finite private algebra is unitary, and given that what any observer can

hold is a public record produced by a specific channel, account exactly for the information that the record does not carry.

The model supplies sharper versions of both sides than a generic quantum system would.

First, within the supplied pointwise-continuous star-automorphism class, unitarity is not an additional assumption. The Stone-converse packet proves that every pointwise-continuous one-parameter star-automorphism group of a unital finite-dimensional private star subalgebra is blockwise unitary conjugation generated by one self-adjoint Hamiltonian per block. The theorem makes no claim about dynamics outside that class.

Second, the record channel is not a generic decoherence map. For every projective partition, the publicization pinching is exactly the uniform average of the 2^k independently signed block reflections. The channel that produces the public record is therefore an explicit uniform mixture of unitaries, with the mixing unitaries named. It is unital; entropy nondecrease for this channel is part of Conjecture 3.1.

The conjectured ledger combines these two facts with the exact thermodynamic theorems. It would identify the information absent from the public record with observer-inaccessible coherences of the ambient state through the formula in Conjecture 3.1.

2 Machine-checked inputs

Each input below is a machine-checked result in the repository, cited by module. Their joint consequence and the precise conjectures are stated in Section 3.

Theorem 2.1 (Forced unitarity; finite Stone packet)**.** *Every continuous one-parameter group of star automorphisms of a unital, finite-dimensional private star subalgebra containing the ambient identity acts by blockwise unitary conjugation. On each Wedderburn block it is generated by a fixed self-adjoint Hamiltonian, unique up to a real scalar.*

Theorem 2.2 (Publicization is a uniform mixture of sign unitaries)**.** *For every supplied k-block projective partition, the pinching map is the uniform average of the 2^k diagonal sign unitaries (`EventAlgebra/RecordMajorization.lean`); averaging only the two global signs does not pinch, so the independence of the signs is load-bearing.*

Theorem 2.3 (Exact data processing and second law)**.** *Every stochastic kernel preserving a faithful stationary reference contracts Kullback–Leibler divergence to that reference (`kl_push_le`), the conditional heat bath obeys the second law (`heatBath_secondLaw`). For finite real-valued distributions and kernels, the integral fluctuation identity holds under its positivity, normalization and stationary-reference hypotheses; the pointwise Crooks identity additionally assumes detailed balance. The proofs are in `Lean/Thermodynamics/FiniteConditionalRepair.lean` and `Lean/Thermodynamics/FluctuationTheorems.lean`.*

Theorem 2.4 (Support boundary for entropy architecture)**.** *The totalized continuous-functional-calculus logarithm sends every projection to zero, so the raw finite trace formula for relative entropy assigns the value zero to an orthogonal pair of density projections whose support inclusion fails (`EventAlgebra/SpectralEntropyBoundary.lean`). A support-aware extended-value layer is therefore mandatory, not stylistic.*

Theorem 2.5 (Conditional rich-fibre regional packet)**.** *The locally pinned fresh-seed source run supplies split-fibre labels at four pairwise-disjoint observer windows. A declared finite adapter constructs a genuinely noncommutative block algebra with a nonunital two-by-two designated-fibre matrix corner at every observer region, elementwise commutation across distinct constructed blocks,*

computed four-window coverage of the declared top algebra, and unique restriction gluing with its reconstruction corollary (`QFT/RichFibreWitness.lean`, `QFT/RichFibreRegionalNet.lean`). The source does not produce the matrix units, state, regional restrictions, or repair, and the corner is neither a tensor factor nor a `TensorSplitReceipt`. A source-attached regional-factor theorem requires an operator algebra and a justified factor or local-channel adapter.

Theorem 2.6 (Finite KMS identity, degenerate inheritance boundary)**.** *The finite algebraic KMS identity holds for Gibbs states of support-graded flows, and the truncation-8 nets inherit KMS structure only degenerately (`QFT/StructuralInheritance.lean`). The nondegenerate continuation on the rich-fibre net is Conjecture 3.6.*

3 Finite conjectures

Throughout, ρ is a finite density matrix on the ambient algebra, $\mathcal{E}$ the pinching of a supplied projective partition, S the support-aware von Neumann entropy, and $D(\cdot \,\|\, \cdot)$ the support-aware relative entropy. Conjectures 3.1–3.6 state the additional finite hypotheses; they do not follow from the preceding theorems.

Conjecture 3.1 (Support-aware entropy core)**.** *There are finite support-aware definitions of $S(\rho)$ and $D(\rho \,\|\, \sigma)$ obeying the basic entropy laws. For pinching they satisfy*

$$S(\mathcal{E}(\rho)) - S(\rho) = D(\rho \,\|\, \mathcal{E}(\rho)) \geq 0,$$

with equality exactly at the public fixed states, where ($\mathcal{E}(\rho) = \rho$), and the sign-unitary mixture is entropy-nondecreasing. This core is the subset of the spectral-information chain required by the ledger; the more general majorization, maximum-entropy, and H-theorem scope is not used here.

Conjecture 3.2 (Finite no-loss statement)**.** *Under the specified ambient repair dynamics the global state evolves by the forced unitary conjugation of the Stone packet. Ordinary entropy of the state is preserved; when both arguments co-evolve under the same unitary, each explicitly named jointly unitarily invariant quantity is also preserved (including support-aware relative entropy and trace distance when those quantities are defined). The statement quantifies over the specified dynamics class; no toy evolution is substituted and no arbitrary coordinate-dependent divergence is covered.*

Conjecture 3.3 (Ledger identity and relocation)**.** *For the specified record channel, the public entropy gain equals $D(\rho \,\|\, \mathcal{E}(\rho))$; the quantity is nonnegative, vanishes exactly at the public fixed states, and is carried entirely by the off-block coherences that no access cut of the specified net exposes. Information is relocated, never destroyed, and the relocation is exactly quantified.*

Conjecture 3.4 (Generalized-second-law-shaped monotone)**.** *The composition of the cap first law with Kullback–Leibler contraction yields one exact monotone combining record entropy and the capacity ledger under repair, with finite constants and a negative control exhibiting failure of the monotone when unitality of the record channel is dropped.*

Conjecture 3.5 (Complementarity receipts)**.** *There is a composition of the generic operational access-cut interface, the supplied rich-fibre data, and a justified operator/local-channel adapter that forms a two-observer ledger in which each accessible algebra determines its own record entropy, the declared block-commutation theorem supplies an algebraic locality receipt, and a glued top observable is uniquely reconstructed within the declared restriction system. No physical no-signalling or refinement-natural interpretation follows before that attachment.*

Conjecture 3.6 (Nondegenerate finite KMS receipt)**.** *The modular data of a restricted state on one genuinely noncommutative rich-fibre region satisfies the finite KMS identity at a capacity-linked inverse temperature, nondegenerately. This receipt supplies a finite KMS input but does not establish structural QFT thermality.*

Conjecture 3.7 (Finite Page-crossover inequality)**.** *For a bipartition of the specified finite net, the exact 2^k sign-unitary average obeys a finite inequality that locates the crossover of accessible-record information without Haar integration.*

Remark 3.8 (Conditional finite implication). Conjectures 3.1–3.6 would imply simultaneous ambient unitarity and a thermal public record, with the inaccessible information equal to the computable functional $D(\rho \,\|\, \mathcal{E}(\rho))$ and localized in observer-inaccessible coherences. The implication supplies no physical black-hole attachment. A source-selected order embedding with calibrated density (a causal-set faithful embedding), manifoldlikeness, and a source-realized clock are additional hypotheses. Any smooth Einstein reading also requires same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification.

4 Seam-count area law on the source family

The layered read law of the conservative source family supplies a finite count that scales with area. Fix a population of constant density n, sites per unit volume, in a bounded convex window $\Omega \subset \mathbb{R}^3$, a read radius a, and the complete-neighbour read law: an event at layer $j+1$ reads every event at layer j within distance a, the same site included. For a plane section $\Sigma = \Omega \cap \{x_1 = P\}$, a crossing read of one layer is an unordered site pair $\{s, t\}$ with $s \neq t$, $\|t - s\| \leq a$, and $s_1 < P < t_1$. Each such pair is realized in every layer by one read in each direction across Σ, from the lower layer to the upper layer; the ordered read count is therefore twice the unordered count, and the unordered count per layer is the primary quantity here. A same-site read never crosses a plane, and a site with $s_1 = P$ belongs to neither side.

Proposition 4.1 (Crossing reads per unit area)**.** *Let $S_k \subset \Omega$ be finite populations with a partition of Ω into cells of common volume $1/n_k$, each cell contained in the ball of radius H_k about its site, with read radii $a_k \to 0$ and $H_k/a_k \to 0$. Let $\Sigma = \Omega \cap \{x_1 = P\}$ have positive area $|\Sigma|$. The number X_k of unordered crossing reads of Σ per layer satisfies*

$$X_k = n_k^2 \,|\Sigma| \int_0^{a_k} \frac{\pi (a_k - u)^2 (2a_k + u)}{3} \,\mathrm{d}u \,(1 + o(1)) = \frac{\pi}{4}\, n_k^2 a_k^4 \,|\Sigma| \,(1 + o(1)). \tag{1}$$

The relative error is bounded by a constant multiple of $H_k/a_k + a_k$, the constant depending on Ω and Σ.

Proof. Write $u(s) = P - s_1$ for the depth of a site below Σ and $C(s) = \{t \in \Omega :\ t_1 > P,\ \|t - s\| \leq a_k\}$, so that $X_k = \sum_{0 < u(s) \leq a_k} |S_k \cap C(s)|$. The cell of a site t lies in $B(t, H_k)$; a cell that meets both $C(s)$ and its complement lies in the $2H_k$-collar of $\partial C(s)$. Hence $|S_k \cap C(s)|$ differs from $n_k \operatorname{vol} C(s)$ by at most n_k times the volume of that collar. The boundary $\partial C(s)$ consists of a spherical cap, a planar disc and pieces of $\partial\Omega$, of total area $O(a_k^2)$, so the error is $O(n_k H_k a_k^2)$ per site. The slab $0 < u(s) \leq a_k$ contains $O(n_k a_k |\Sigma|)$ sites, which gives a total error $O(n_k^2 H_k a_k^3 |\Sigma|)$, of relative order H_k/a_k.

The function $f(s) = n_k \operatorname{vol} C(s)$ on the slab is bounded by $\frac{2\pi}{3} n_k a_k^3$ and is Lipschitz with constant $O(n_k a_k^2)$: its depth derivative is $-\pi n_k (a_k^2 - u^2)$ and a lateral displacement changes $C(s)$ through the faces of Ω by a set of measure at most πa_k^2 per unit displacement. The same cell argument,

applied to the slab, bounds $|\sum_{0<u(s)\le a_k} f(s) - n_k \int_{0<u\le a_k} f\,\mathrm{d}^3 s|$ by n_k times the Lipschitz constant times H_k times the slab volume, plus the contribution of the sites within H_k of the two slab faces; both are $O(n_k^2 H_k a_k^3 |\Sigma|)$, again of relative order H_k/a_k.

Finally, for s at lateral distance more than a_k from the relative boundary of Σ in the plane, $C(s)$ is the untruncated spherical cap of $B(s, a_k)$ beyond height u, with volume

$$\int_u^{a_k} \pi(a_k^2 - z^2)\,\mathrm{d}z = \frac{\pi(a_k - u)^2(2a_k + u)}{3}.$$

The truncated sites occupy a lateral collar of area $O(a_k)$ relative to $|\Sigma|$, so the integral over the slab equals $|\Sigma| \int_0^{a_k} \pi(a_k - u)^2(2a_k + u)/3\,\mathrm{d}u$ up to a relative error $O(a_k)$. The depth integral is $\pi a_k^4/4$. □

Corollary 4.2 (Golden-family value)**.** *For the golden source population S_q of q^3 sites in $\Omega = [0, L]^3$, $L = 2/\sqrt{\varphi + 2}$, with $n_q = q^3/L^3$, $a_q = L/\sqrt{q}$ and cell assignment error $H_q = 2\sqrt{3}L/q$, one has $n_q^2 a_q^4 = q^4/L^2$, and the unordered crossing count per layer through a full cross-section $\Sigma = \Omega \cap \{x_i = P\}$ is*

$$X_q = \frac{\pi q^4}{4}(1 + O(q^{-1/2})). \tag{2}$$

For the mid-plane $P = L/2$ the continuum count with the caps truncated by the cube is

$$n_q^2 \int_{s\in\Omega,\, s_1<L/2} \mathrm{vol}\{t \in \Omega :\ t_1 > L/2,\ \|t - s\| \le a_q\}\,\mathrm{d}^3 s = q^4 \left(\frac{\pi}{4} - \frac{8}{15\sqrt{q}} + \frac{1}{12\,q}\right). \tag{3}$$

Proof. Here $H_q/a_q = 2\sqrt{3}\,q^{-1/2}$ and $a_q/L = q^{-1/2}$, so Proposition 4.1 gives (2). For (3) substitute $d = t - s$. For $0 < d_1 \le a_q \le L/2$ and $|d_2|, |d_3| \le L$, the set of $s \in \Omega$ with $s + d \in \Omega$ and $s_1 < L/2 < s_1 + d_1$ has volume $d_1(L - |d_2|)(L - |d_3|)$. Over the half ball $d_1 > 0$, $\|d\| \le a_q$, the moments are $\int d_1 = \pi a_q^4/4$, $\int d_1|d_2| = 4a_q^5/15$ and $\int d_1|d_2||d_3| = a_q^6/12$. Hence the count equals $n_q^2 a_q^4(\pi L^2/4 - 8La_q/15 + a_q^2/12)$, which is (3) after $n_q^2 a_q^4 = q^4/L^2$ and $a_q = L/\sqrt{q}$. □

Proposition 4.3 (Area law and the Planck-area dictionary)**.** *Declare that each crossing read carries one nat of cut entropy across Σ per layer. In the limit of Proposition 4.1 the cut entropy is the area law*

$$S_{\mathrm{cut}}(\Sigma) = \frac{\pi}{4}\, n^2 a^4\, |\Sigma|. \tag{4}$$

Identifying (4) with $|\Sigma|/(4\ell_P^2)$ fixes

$$n^2 a^4 = \frac{1}{\pi \ell_P^2}, \qquad \ell_P = \frac{1}{\sqrt{\pi}\, n\, a^2}, \tag{5}$$

and on the golden family $\ell_P = L/(\sqrt{\pi}\, q^2)$.

Proof. Equation (4) is (1) with one nat per read. Equating $(\pi/4)n^2a^4$ with $1/(4\ell_P^2)$ and solving for ℓ_P^2 gives (5); with $n^2a^4 = q^4/L^2$ this reads $\ell_P^2 = L^2/(\pi q^4)$. □

Remark 4.4 (What is not claimed). Equation (5) is a calibration of the read scale by the area coefficient, in the same way that the layer duration a/c is a calibration of the clock. No horizon is constructed: Σ is a plane section of the source window, with the same read law on both sides. The nat-per-read reading is a declared dictionary, without a derivation from the record channel of the finite ledger. No temperature, no Hawking flux and no physical value of ℓ_P follow; L is the side of the source window, and the family fixes only the ratio $\ell_P/L = 1/(\sqrt{\pi}\, q^2)$. Dou and Sorkin [7]

count causal links across a horizon in a Poisson-sprinkled causal set and obtain an area law whose coefficient depends on the sprinkling; here the population is deterministic, the count is exact at every finite q, and the coefficient $\pi/4$ in (1) is the exact limit.

The depth integral in (1), the value $(\pi/4)n^2a^4$, the golden identity $n_q^2 a_q^4 = q^4/L^2$ with the cross-section value $\pi q^4/4$, and the dictionary (5) with its golden value $\ell_P^2 = L^2/(\pi q^4)$ are machine-checked in `Geometry/CrossingReadAreaLaw.lean` (`capVolume_integral`, `crossingDensity_eq`, `golden_crossing_count`, `planck_area_of_area_law`, `golden_planck_length_sq`) with the standard axioms only. The limit statement of Proposition 4.1 is analytic and is proved above.

Finite receipt. The exact crossing counts of the golden family at $q = 5, 8, 13, 21, 34, 55$ are recorded with an independent verifier. The population and the neighbour relation are rebuilt from the definitions, the relation agrees edge for edge with the causal-poset evidence package at every level, every side decision is exact in $\mathbb{Q}(\varphi)$, and no site lies on any of the five planes $x_1 = L/2$, $x_1 = L/4$, $x_1 = 3L/4$, $x_2 = L/2$, $x_3 = L/2$. Table 1 lists the mid-plane counts. The counts across $x_2 = L/2$ and $x_3 = L/2$ equal the $x_1 = L/2$ count at every level, as the product population is symmetric under coordinate permutations. At $q = 34$ the mid-plane count is 907 080 against the box-corrected value 930 602.0, ratio 0.975, and against the infinite-plane value 1 049 555.8, ratio 0.864; at $q = 55$ the count is 6 496 004 against 6 542 684.6, ratio 0.993, and against 7 186 884.1, ratio 0.904. The ratio to the box-corrected value is closer to one than the ratio to the infinite-plane value at every level; it lies within 0.13 of one at every level and within 0.026 of one for $q \geq 21$, while the ratio to the infinite-plane value rises from 0.658 to 0.904. The residual at fixed q is the finite population itself: the two sides of the mid-plane hold 84 700 and 81 675 sites at $q = 55$, and the mean number of neighbours of a site, 1459.3 at $q = 55$, compares with the cube mean 1461.1 of $n_q \operatorname{vol}(B(s, a_q) \cap \Omega)$ and with the untruncated ball value 1708.6.

q	count	$\pi q^4/4$	ratio	box-corrected	ratio
5	323	490.9	0.658	352.2	0.917
8	2 164	3 217.0	0.673	2 487.3	0.870
13	17 754	22 431.8	0.791	18 390.1	0.965
21	129 830	152 745.0	0.850	130 882.5	0.992
34	907 080	1 049 555.8	0.864	930 602.0	0.975
55	6 496 004	7 186 884.1	0.904	6 542 684.6	0.993

Table 1: Unordered crossing reads per layer across the mid-plane $x_1 = L/2$ of the golden family, the infinite-plane value $\pi q^4/4$, and the box-corrected value (3), with the ratios of the count to each.

5 Relation to standard approaches

The Page analysis, the Hayden–Preskill protocol, and island computations concern continuum or holographic systems with gravitational dynamics. The finite ledger formulates their information question using separately proved unitary dynamics and a specified record channel. Its proposed entropy and thermality composition remains conditional on Conjectures 3.1–3.6. Within the supplied automorphism class, the Stone converse gives unitary implementation; the public channel is the explicit sign-unitary mixture. A physical black-hole interpretation additionally requires the spacetime, horizon and stress attachments stated above.

6 Observables discipline

The finite ledger implies no observable black-hole signature. A proposed observable claim derived from the ledger, such as saturation-regime relaxation structure or record-capacity signatures in compact-object transients, requires a prospectively fixed clock, frame, scale, and readout contract under immutable custody before any data comparison.

A conditional instrument template is defined for the integer-k ringdown transition comb. It is not preregistered or frozen and therefore supports no observable claim. The transition rule is an imported continuation premise, not a source-derived consequence of the finite ledger. For an emission it assumes positive integer record dimensions satisfying $d_{\text{before}} = k d_{\text{after}}$, so k must divide the initial dimension. The signed black-hole entropy change is $\ln d_{\text{after}} - \ln d_{\text{before}} = -\ln k$; the positive entropy loss entering the emitted-energy formula is $\ln k$. For positive remnant mass and subextremal spin $|\chi| < 1$, conditional on that rule, candidate features above the rotation line sit at $x_k = \ln k/(8\pi)$, with $x = (GM/(c^3 g(\chi)))(\omega - m\Omega_H)$ and $g(\chi) = 2\sqrt{1-\chi^2}/(1+\sqrt{1-\chi^2})$. Their offset-subtracted ratios equal $\ln k_a/\ln k_b$ under a common scale and offset for a denominator tooth $k_b \geq 2$, whose offset-subtracted value is proved nonzero. At $|\chi| = 1$, $g(\chi) = 0$: the candidate teeth collapse to the rotation line, and the rescaled coordinate and these ratios are undefined. Observed frequencies must use the detector-frame remnant mass $M_{\text{det}} = (1+z)M_{\text{source}}$ in both terms; the ratio is redshift-free only with this consistent frame convention. The tooth relation uses the leading small-transition Kerr first law; finite-step background corrections are not derived by this template.

The declared linewidth-to-spacing ratio is $64\pi^2 p_0/(a g(\chi)^2 \ln k)$, which is mass-independent but spin-dependent; a constant Page coefficient is not controlled near extremal spin. The declared $(k-1)/k$ KMS factor is a within-line net-response factor, not a cross-k transition probability or prior. The algebraic invariance, divisibility and entropy-sign boundary, ladder bracketings, producer, independent byte-identical verifier, and tests are target-blind. No event likelihood has been evaluated. Published posterior samples alone are neither a strain likelihood nor a model evidence. This template is not anchorable until a source-derived transition, common detector likelihood or sufficient likelihood product, detector-frame nuisance model, normalized integer prior or explicit finite submodel, event selection, trials, greybody normalization, no-comb reference, and convergence rule are fixed. The computed $2 \leq k \leq 12$ ladder is only a finite candidate submodel; excluding it would not reject the unrestricted integer family.

7 Verification surface

The proved theorems cited in Section 2 carry `#print axioms` receipts with only the standard axioms, no admissions, and no native-evaluation shortcut, in the modules named inline. The support-aware entropy statements use the general spectral-information definitions. Conjectures 3.1–3.6 are not proved by those receipts.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Bekenstein–Mukhanov area quantization, Kerr ringdown spectroscopy, and quasinormal-mode templates [8, 10, 9].

- Dou and Sorkin (2003), black-hole entropy as causal links crossing the horizon of a Poisson-sprinkled causal set [7].

References

[1] S. W. Hawking, *Particle creation by black holes*, Commun. Math. Phys. 43 (1975) 199–220.

[2] S. W. Hawking, *Breakdown of predictability in gravitational collapse*, Phys. Rev. D 14 (1976) 2460–2473.

[3] D. N. Page, *Average entropy of a subsystem*, Phys. Rev. Lett. 71 (1993) 1291–1294.

[4] P. Hayden and J. Preskill, *Black holes as mirrors: quantum information in random subsystems*, JHEP 09 (2007) 120.

[5] A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, and A. Tajdini, *The entropy of Hawking radiation*, Rev. Mod. Phys. 93 (2021) 035002.

[6] G. Lindblad, *Completely positive maps and entropy inequalities*, Commun. Math. Phys. 40 (1975) 147–151.

[7] D. Dou and R. D. Sorkin, *Black-hole entropy as causal links*, Found. Phys. 33 (2003) 279–296.

[8] Bekenstein, J. D., and Mukhanov, V. F. (1995). "Spectroscopy of the quantum black hole." *Physics Letters B* **360**, 7–12. https://doi.org/10.1016/0370-2693(95)01148-J

[9] Berti, E., Cardoso, V., and Starinets, A. O. (2009). "Quasinormal modes of black holes and black branes." *Classical and Quantum Gravity* **26**, 163001. https://doi.org/10.1088/0264-9381/26/16/163001

[10] Dreyer, O., Kelly, B., Krishnan, B., Finn, L. S., Garrison, D., and Lopez-Aleman, R. (2004). "Black-hole spectroscopy: testing general relativity through gravitational-wave observations." *Classical and Quantum Gravity* **21**, 787–803. https://doi.org/10.1088/0264-9381/21/4/003

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.